

RESEARCH PAPER 1

Revolutionizing Emergency Medical Services: A Review of Ambulance Technology Advancements and Service Enhancement Strategies

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Abstract:

This paper addresses the challenges faced by current emergency ambulance systems, focusing on improving service efficiency and patient outcomes. Analyzing various studies, we explore the demand for emergency services, including factors such as traffic congestion and inappropriate use of ambulances. Additionally, queuing theory models are examined to determine optimal fleet sizes for meeting service standards. The review also highlights advancements in ambulance technologies, emphasizing the potential for enhancing emergency medical services (EMS) effectiveness and efficiency. Through a systematic analysis of ambulance technologies from the past decade, this paper identifies trends, growth patterns, and future directions for EMS technology. The findings underscore the importance of investing in innovative ambulance technologies to improve patient care and survival rates. Recommendations for developing an app-based, 24/7 emergency service connected to hospitals are proposed to address current EMS challenges effectively.